

Cauchy-Schwarz Inequality

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Equality holds iff x and y are linearly dependent

- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Continuity of inner product
- Orthogonality characterization
- Best approximation theory